

Your grade: 100%Your latest: **100%** • Your highest: **100%** • To pass you need at least 75%. We keep your highest score.[Next item →](#)

1. What types of errors might you encounter while debugging code? Select three answers.

1 / 1 point

☒ Exceptions

Correct

Syntax errors, logic errors, and exceptions are all types of errors you might encounter while debugging code. Syntax errors involve invalid usage of the Python language. Logic errors may not cause error messages, but they produce unintended results. Exceptions happen when the program does not know how to execute code even though it is syntactically correct.

☐ Iteratives☒ Syntax errors

Correct

Syntax errors, logic errors, and exceptions are all types of errors you might encounter while debugging code. Syntax errors involve invalid usage of the Python language. Logic errors may not cause error messages, but they produce unintended results. Exceptions happen when the program does not know how to execute code even though it is syntactically correct.

☒ Logic errors

Correct

Syntax errors, logic errors, and exceptions are all types of errors you might encounter while debugging code. Syntax errors involve invalid usage of the Python language. Logic errors may not cause error messages, but they produce unintended results. Exceptions happen when the program does not know how to execute code even though it is syntactically correct.

2. The purpose of this code is to indicate whether a particular operating system needs to be updated. However, it contains a syntax error. Run this code, analyze its output, and then debug it. (If you want to undo your changes to the code, you can click the Reset button.)

1 / 1 point

```
1 operating_system = "OS 2"
2 if operating_system == "OS 1":
3     print("updates needed")
4 elif operating_system == "OS 2":
5     print("no updates needed")
```

[Run](#)[Reset](#)

Based on what you discover, how can you fix the error?

- ☒ Change the keyword **elif** to **elif**.
- ☐ Use single equals signs (=) and not double equals signs (==).
- ☐ Remove all colons (:).
- ☐ Indent the **elif** statement.

Correct

When you run this code, the error message can help you identify the syntax error and the line number where it occurs. Changing the keyword **elif** to **elif** will fix the error. Syntax errors involve invalid usage of the Python language, such as misspelling a keyword. The correct spelling for the keyword needed before the condition **operating_system == "OS 2"** is **elif**.

3. You have written code that assigns security incident tickets to the appropriate cybersecurity team based on its priority level. If the priority level is 1, it should get forwarded to Team A. If the priority level is 2, it should get forwarded to Team B. When testing your code, you notice that an incident with priority level 2 is forwarded to Team A instead of Team B. What type of error is this?

1 / 1 point

- ☒ Logic error
- ☐ Exception
- ☐ Name error
- ☐ Syntax error

Correct

This is a logic error. Logic errors are errors that result when the logic used in code produces unintended results. In this situation, because the security incident ticket is forwarded to the wrong team, there is an unintended result.



- ☐ Name error
- ☐ Syntax error

**Correct**

This is a logic error. Logic errors are errors that result when the logic used in code produces unintended results. In this situation, because the security incident ticket is forwarded to the wrong team, there is an unintended result.

4. You have written code that uses a search algorithm to find an employee's IP address. When testing your code, an error message indicates that an unknown index is being accessed. What type of error is this?

1 / 1 point

- ☒ Exception
- ☐ Logic error
- ☐ Syntax error
- ☐ Iterative

**Correct**

This is an exception. Exceptions occur when Python does not know how to execute code even though it is syntactically correct. This happens if you ask Python to access an index that does not exist.